

Problem Statement and Goals

Cut Edge Weight: Data Separability Measurement

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Table 1: Revision History

Date	Developer(s)	Change
2023-01-20	Lesley Wheat	Creation

1 Problem Statement

Classification is a common machine learning problem and there are many techniques for solving classification problems. Unfortunately, not all classification problems are easy and it can be difficult to explain why classifiers perform poorly on certain datasets. Data complexity has been defined as a characteristic of the data that relates to the difficulty of the classification problem. [1] To further refine this definition, we consider data complexity to be represented by aspects of the data that introduce difficulty for the classifiers.

One of these aspects is separability, an intrinsic characteristic that describes how the data points from different classes mix together. [2] A classic measurement for separability is Fraction of Points on Class Boundary. Based on the work of Friedman and Rafsky [3] and documented by Basu and Ho[4], this measurement constructs a minimum spanning tree over the dataset then calculates the number of points with a connection to another class over all points.

The Cut Edge Weight measurement [5] is a variation of the Fraction of Points on Class Boundary, proposed by Zighed et al. The algorithm for the normalized measurement begins by constructing a Toussaint’s Relative Neighbourhood Graph [6] over the entire dataset then computing the number of edges which connect different classes over the total number of edges. This measurement is a viable way of measuring separability because it has a high coefficient of determination (R^2) with the mean error rate of several classifiers. This algorithm does not currently have an available open source implementation, unlike others listed in [4].

1.1 Problem

The problem is the computation of the Cut Edge Weight measurement over a dataset.

1.2 Inputs and Outputs

1.2.1 Inputs

The program should receive the dataset as an input. The dataset should contain samples with features (in machine learning, a feature is an individual measurable property or characteristic [7]). Each sample should have the same number of features. Each feature should have a numerical value.

1.2.2 Outputs

The program should output the value of the normalized Cut Edge Weight measurement for the input dataset as a floating point number.

1.3 Stakeholders

The stakeholders are machine learning researchers.

1.4 Environment

The program should run on a Windows 11 operating system. The program should run on a laptop and should not require a GPU.

2 Goals

The goal of the software is to calculate the value of the normalized Cut Edge Weight Statistic as defined in [5] using a Relative Neighbourhood Graph with no weighting.

3 Stretch Goals

Stretch goals include:

- The option to weigh the edge calculation using the distance measurement of $(1 + d_{ij})^{-1}$ where d_{ij} is the Euclidean distance between points i and j .
- The option to weigh the edge calculation using rank so that the weight is r_i^{-1} . Where r_j is calculated as the rank of vertex j among the neighbours of vertex i .
- Generate the p-value of the statistic as defined in [5].

- Give the user the option to run the algorithm using a Gabriel Graph instead of Relative Neighbourhood Graph.
- Give the user the option to run the algorithm using a Minimal Spanning Tree instead of Relative Neighbourhood Graph.
- Give the user the option to output the statistic without normalization.
- Package the code in a way accessible to the public.

References

- [1] J. Luengo and F. Herrera, “An automatic extraction method of the domains of competence for learning classifiers using data complexity measures,” *Knowledge and Information Systems*, vol. 42, no. 1, pp. 147–180, Jan. 2015. [Online]. Available: <http://link.springer.com/10.1007/s10115-013-0700-4>
- [2] S. Guan and M. Loew, “A Novel Intrinsic Measure of Data Separability,” Sep. 2021, arXiv:2109.05180 [cs, math, stat]. [Online]. Available: <http://arxiv.org/abs/2109.05180>
- [3] J. H. Friedman and L. Rafsky, “Multivariate Generalizations of the Wald-Wolfowitz and Smirnov Two-Sample Tests,” *The Annals of Statistics*, vol. 7, no. 4, pp. 697–717, 1979. [Online]. Available: <https://www.jstor.org/stable/2958919>
- [4] T. K. Ho and M. Basu, “Complexity measures of supervised classification problems,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 24, no. 3, pp. 289–300, Mar. 2002, conference Name: IEEE Transactions on Pattern Analysis and Machine Intelligence.
- [5] D. A. Zighed, S. Lallich, and F. Muhlenbach, “A statistical approach to class separability,” *Applied Stochastic Models in Business and Industry*, vol. 21, no. 2, pp. 187–197, Mar. 2005. [Online]. Available: <https://onlinelibrary.wiley.com/doi/10.1002/asmb.532>
- [6] G. T. Toussaint, “The relative neighbourhood graph of a finite planar set,” *Pattern Recognition*, vol. 12, no. 4, pp. 261–268, Jan. 1980. [Online]. Available: <https://linkinghub.elsevier.com/retrieve/pii/0031320380900667>
- [7] C. M. Bishop, *Pattern recognition and machine learning*, ser. Information science and statistics. New York: Springer, 2006.